

When crop rankings do not determine land allocations: set-valued reversal under operational and downside-risk constraints

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Abstract

Crop rankings reduce heterogeneous evidence to an order, whereas land allocation is a cardinal decision made under joint uncertainty and operational restrictions. We formalize this gap with a set-valued, multi-crop programme that maximizes expected profit subject to a polyhedral feasible set and loss-based conditional value at risk. The theory distinguishes possible, universal and selected reversal, proves that rankings alone cannot identify acreage, and shows that operational feasibility can force reversal even when downside risk is inactive. In preregistered simulations, operational interventions convert a corn-only solution into soybean-majority allocations in all treated E2 cells; all 24/24 family-wise intervals meet their precision targets. A separate E6 experiment shows that the interaction between information and flexibility can be positive (3.258, 95% interval 2.788 to 3.728), exactly null, or substitutive, establishing actionability without universal complementarity. Four other experiments fail their registered precision gates and remain adverse evidence. In admitted US state data, concurrent operating-margin-acreage inversion intensity is 0.597 (state-cluster 95% interval 0.494 to 0.697), but all four lagged share-change intervals include zero and national results depend on aggregation. Thus ranking reversal is theoretically coherent and reproducible under explicit mechanisms, while observed acreage disagreement alone does not identify private constraints, downside-risk preferences, dependence, causality or welfare.

1 Introduction

Crop prediction systems increasingly transform weather, soil, management and remote-sensing data into yield forecasts, suitability scores and crop rankings [Liakos et al., 2018, van Klompenburg et al., 2020]. These outputs answer an ordinal question—which crop scores higher—but not the operational question of how much land to assign to each crop. Acreage decisions require cardinal margins, a joint uncertainty model, land and liquidity limits, rotations, contracts, bounds and a rule for resolving multiple optima. Treating a top-ranked crop as the crop that should receive the most land therefore crosses a consequential boundary from prediction to prescription.

That boundary matters because the two objects have different invariances. A ranking survives every strictly increasing transformation of its scores; a proportional allocation generally does not. A ranking also contains no information about the geometry of a farm’s feasible set, the tails of portfolio profit, or the action selected from a non-singleton optimal face. The apparent simplicity of an ordered crop list can consequently conceal several distinct explanations for the same acreage pattern: cardinal margins may disagree with suitability scores, a constraint may directly displace the leader, a risk limit may alter portfolio exposure, or optimizer multiplicity may make the reported allocation selection-dependent.

Crop-planning research has long represented land use as a constrained portfolio problem, incorporating resource limits, stochastic prices and yields, rotations, robustness and downside risk [Filippi et al., 2017, Boyabatlı et al., 2019, Benini et al., 2023, Randall et al., 2022, 2024]. In parallel, prescriptive analytics and decision-focused learning distinguish predictive accuracy from downstream decision

quality [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022, Wilder et al., 2019, Mandi et al., 2023]. Yet the interpretation of rank–allocation disagreement is often under-specified: a difference between an ordered score and an acreage vector does not by itself reveal whether a decision is infeasible, risk-sensitive, inefficient or even unique.

Here we reconstruct the ranking-to-allocation argument as a closed evidence chain (Fig. 1). We retain the original multi-crop stochastic architecture but make its objects and identification limits explicit. The model maximizes expected profit over a polyhedral operational set under loss-CVaR, using the atom-safe representation of Rockafellar and Uryasev [2000, 2002]. The optimum is a set. Reversal can therefore be possible at some optimum, universal at every optimum, or selected under a declared deterministic rule. This distinction is not semantic: exact constructions show that margin, feasibility, risk and multiplicity can produce visually similar allocations for different mathematical reasons.

The theoretical results deliver three mechanisms and two boundaries. First, ordinal information alone cannot identify cardinal acreage. Second, operational constraints can force universal reversal without any risk channel. Third, the complete Karush–Kuhn–Tucker identity decomposes local pressure into margin, tail-risk, budget, shared-constraint and bound terms, while refusing the invalid shortcut from marginal conditions to acreage order. Dependence comparisons require a distributional order over portfolio losses rather than a scalar tail-dependence coefficient, and reversal regions need not collapse to a unique threshold. Information has non-negative value when it can be ignored, but its interaction with flexibility has no universal sign.

We test these propositions with six preregistered confirmatory experiments. Only E2 and E6 meet every experiment-level precision criterion and are promoted. In E2, operational interventions move the solution from 1.000 corn and zero soybeans to either 0.450 corn and 0.550 soybeans or 0.318 corn and 0.682 soybeans; possible, universal and selected reversal coincide in every treated cell. In E6, information and flexibility are complementary only when flexibility unlocks state-contingent specialization. A robust fallback instead makes them substitutes, and a dominated option yields an exact-zero interaction. E1, E3, E4 and E5 reach their stopping ceilings without passing every registered interval and are retained only as adverse evidence.

The empirical analysis asks a narrower, data-admissible question. It compares four transparent score definitions with planted-acreage shares in 77 state-years from 26 US states over 2022–2024. Concurrent operating-margin inversion intensity is 0.597, but temporal and aggregation checks constrain interpretation: every lagged top-minus-other share-change interval includes zero, the acreage leader changes in only 5 of 51 state transitions, national operating-margin inversion is 0.000, and national total-cost-margin inversion is 0.667. These patterns establish descriptive heterogeneity, not that observed acreage solves the model.

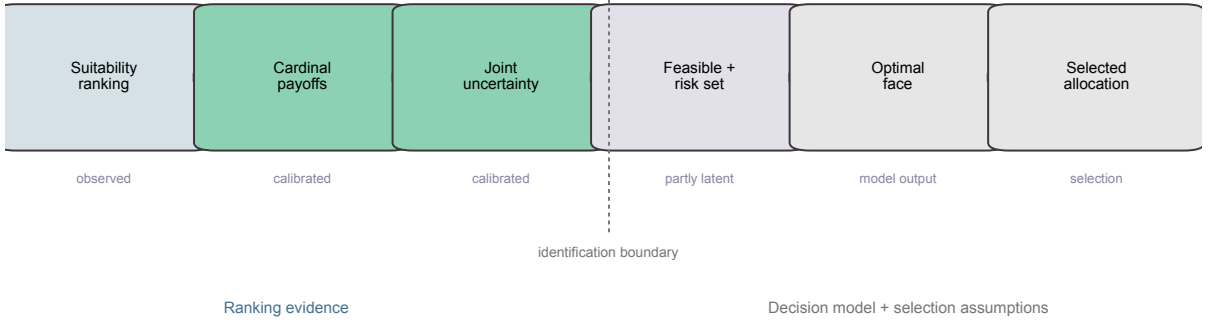
The contribution is therefore both positive and disciplined. Ranking reversal exists as a formally defined relationship between a ranking and an optimal set. Operational displacement and information actionability can be identified in controlled, precision-qualified experiments. Observed rank–acreage inversions are visible and definition-sensitive in aggregate data. But the evidence does not convert public acreage records into estimates of private objectives, constraints, CVaR limits, copula mechanisms, welfare or causality. By preserving those boundaries, the paper turns a broad ranking-reversal intuition into a reproducible mechanism framework.

2 Related literature

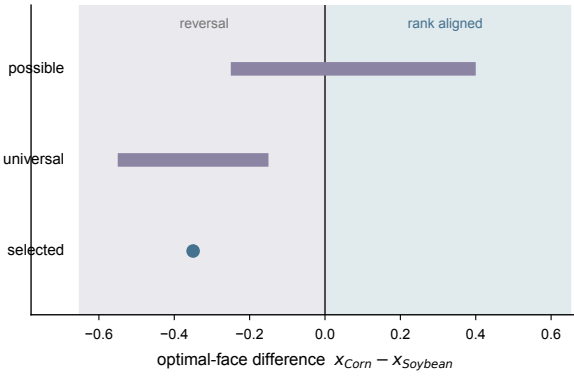
Prediction and prescription in agriculture. Agricultural machine learning commonly predicts yields or classifies agronomic outcomes from weather, soil, sensor and remote-sensing inputs [Liakos et al., 2018, van Klompenburg et al., 2020]. Such models can support a ranking, but they do not

a

From ordinal rankings to identified allocation claims



b



c

- 1 Is the ranking ordinal or cardinal?
 - 2 Are joint returns identified?
 - 3 Which constraints are observed?
 - 4 Is the optimum set-valued?
 - 5 What selects an acreage vector?
- Observed acreage alone cannot recover private objectives or constraints.

Figure 1: Identification architecture and reversal definitions. Rankings become allocation claims only after cardinal payoffs, joint uncertainty, feasibility and a selection rule are added. Possible and universal reversal are properties of the full optimal face; selected reversal depends on the reported optimizer. Colours follow the frozen Stage II palette and are redundant with labels and position.

encode the feasible set or objective required for acreage prescription. The broader prescriptive-analytics literature evaluates predictions through downstream decisions and regret [Bertsimas and Kallus, 2020, Elmachoub and Grigas, 2022], while decision-focused learning differentiates or searches through optimization structure [Wilder et al., 2019, Mandi et al., 2023]. Our analysis does not propose a new learning algorithm. It instead isolates the identification gap that must be resolved before a prediction can be interpreted as a land allocation.

Constrained crop planning. The closest included crop-planning studies optimize crop mix under stochastic returns, rotations or climate uncertainty. Filippi et al. [2017] combine realistic crop-selection constraints with a CVaR criterion; Boyabath et al. [2019] link allocation and rotation benefits dynamically; Benini et al. [2023] formulate multi-period rotation and diversification constraints; and Randall et al. [2022] study robustness across climate projections. The systematic review by Randall et al. [2024] documents the diversity of optimization approaches in the field. These studies motivate a multi-crop operational set, but they do not equate ordinal recommendations with the entire optimal solution set.

Downside risk and dependence. Loss-CVaR is convex and admits a finite-scenario auxiliary representation that remains valid with atoms [Rockafellar and Uryasev, 2000, 2002]. Copula models

can separate marginal distributions from dependence, but tail properties vary by family [Demarta and McNeil, 2005, Ansari and Rockel, 2024]. We consequently condition any dependence comparison on a named family and a verified ordering of portfolio-loss distributions. This is stricter than treating a lower-tail coefficient as a universal mechanism.

Information and operational flexibility. Stochastic programming separates here-and-now choices from actions taken after information arrives [Dantzig, 1955]. Information has operational value only when it changes feasible decisions [Keisler, 2004, Merkhofer, 1975]; the value of flexible resources likewise depends on margins, costs and the complete uncertainty environment [Van Mieghem, 1998]. These results support non-negativity and weak action-set monotonicity, but not general strict complementarity between signal precision and flexibility.

Relative to this verified set, the present study differs by jointly comparing an ordinal crop ranking with an entire constrained loss-CVaR optimal set, separating possible, universal and selected reversals, and pairing that theory with a claim-bounded descriptive empirical audit. This is a bounded distinction, not a claim to be the first study of crop allocation, CVaR or prediction-informed decisions.

3 Integrated multi-crop stochastic model

Let crops be indexed by $i = 1, \dots, n$ and let x_i denote acreage. Random per-acre profit is $\tilde{\pi}_i = \tilde{p}_i \tilde{y}_i - c_i$, where price and yield may be dependent within and across crops and baseline cost c_i is deterministic. Portfolio profit and loss are

$$\Pi(x, \omega) = \sum_i x_i \tilde{\pi}_i(\omega), \quad L(x, \omega) = -\Pi(x, \omega). \quad (1)$$

Returns are linear in acreage. Fixed setup costs, endogenous prices, crop interactions and multi-period state variables are outside the baseline and would require a different feasible set or objective.

Operational feasibility is represented by the non-empty polytope

$$X = \{x \in \mathbb{R}^n : \ell \leq x \leq u, \mathbf{1}^\top x \leq A, c^\top x \leq B, Gx \leq g\}. \quad (2)$$

The rows of $Gx \leq g$ accommodate rotation, contract, agronomic and other shared restrictions. Land is an inequality because idling can be optimal or forced; full use requires an admissible profitable expansion. In the experiments these constraints are controlled interventions. They are not estimates of private farm restrictions.

For confidence level $\alpha \in (0, 1)$, loss-CVaR is

$$r_\alpha(x) = \text{CVaR}_\alpha[L(x)] = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1 - \alpha} \mathbb{E}[(L(x) - v)_+] \right\}. \quad (3)$$

The canonical problem maximizes expected profit $\mu^\top x$, where $\mu_i = \mathbb{E}[\tilde{\pi}_i]$, subject to $x \in X$ and $r_\alpha(x) \leq \kappa$. The sign convention protects against large losses. For finite scenarios s with positive weights w_s , a free v and non-negative excess-loss variables q_s yield a linear programme.

Marginal profit distributions F_i and a valid n -copula C_θ define the joint law. Pairwise lower-tail dependence is a diagnostic when its limit exists, but does not identify a copula, a conditional tail mean or an allocation. Named Gaussian, Student- t , Clayton and empirical-copula laws are therefore compared as complete scenario-generating models, never as if a single scalar ordered all portfolio losses [Demarta and McNeil, 2005, Ansari and Rockel, 2024].

A suitability vector s induces a weak order, with ties forming equivalence classes. Proportional and winner-take-all rules are benchmark mappings, not optima by definition. Let

$$S_\kappa = \arg \max \{ \mu^\top x : x \in X, r_\alpha(x) \leq \kappa \}. \quad (4)$$

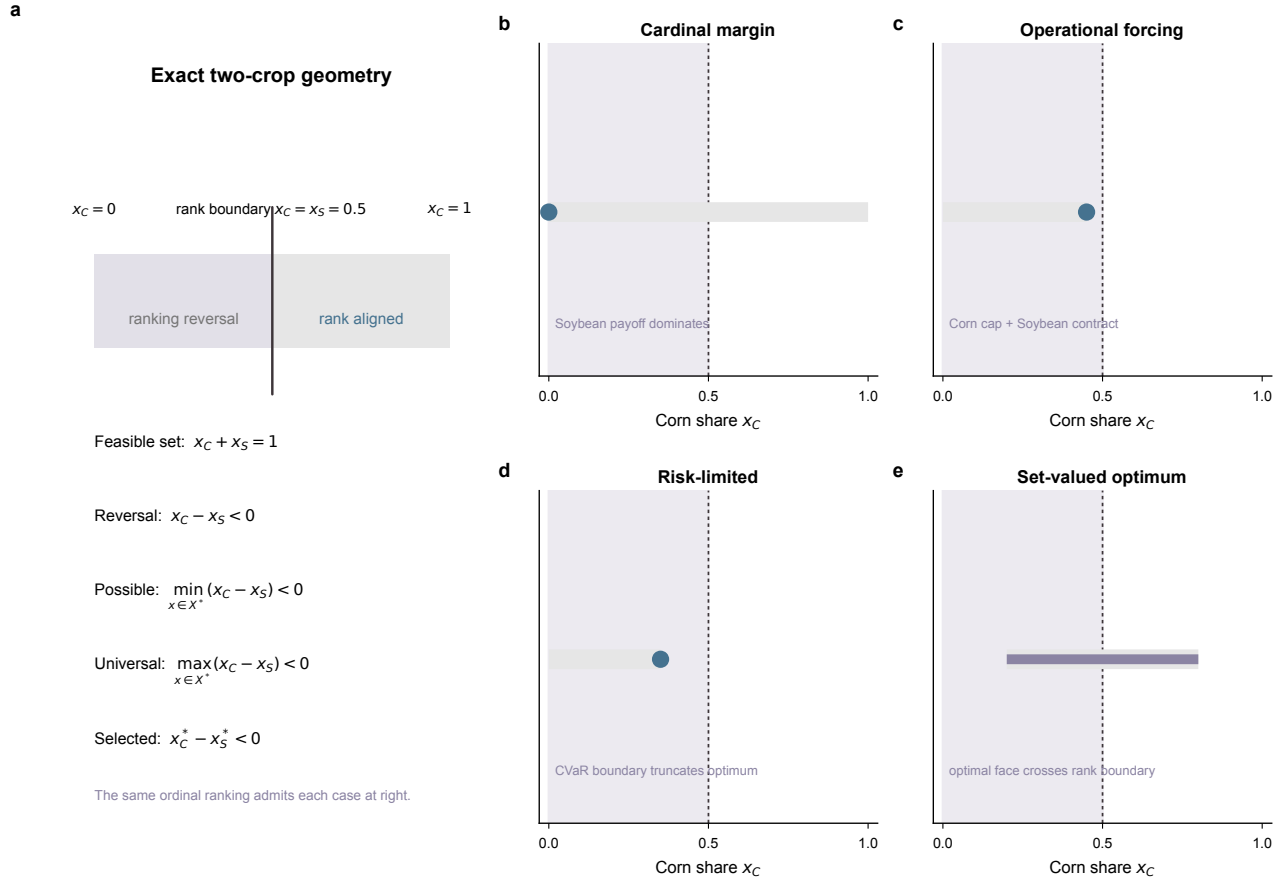


Figure 2: Exact geometry of distinct ranking-reversal mechanisms. Two-crop constructions distinguish cardinal-margin, operational, risk-limited and set-valued mechanisms. Shading marks the corn–soybean reversal half-space. The cases are analytic constructions, not empirical estimates.

For a strict pair $s_i > s_j$ and tolerance τ_x , reversal at x means $x_j - x_i > \tau_x$. It is *possible* if this holds at some $x \in S_\kappa$, *universal* if it holds at every $x \in S_\kappa$, and *selected* if it holds for a declared deterministic selection. Strong and top-rank reversal add zero-allocation and highest-tie-class conditions. This set-valued vocabulary separates properties of the decision problem from properties of one solver-returned vertex.

Figure 2 locates four exact two-crop cases in the same acreage simplex. An expected-margin case reverses the external score because cardinal profit margins point elsewhere. Operational bounds can place the entire feasible set in the reversal half-space. A binding risk limit can move the optimal point without implying monotone crop-wise comparative statics. A flat optimal face can contain both reversing and non-reversing points, so selected reversal depends on the selection rule even though possible and universal reversal remain well defined.

4 Structural results and mechanism decomposition

The first result is an identification theorem: a weak or strict suitability order alone does not determine a cardinal acreage vector. Strictly increasing transformations preserve the order but can change a proportional mapping, and neither transformation nor ranking supplies the objective, feasible set, risk

convention or optimizer selection. Ordinal insufficiency is therefore structural rather than a numerical failure.

Existence and tractability follow under conventional conditions. If X is non-empty and compact, per-acre profits are integrable, and the risk-feasible subset is non-empty, an optimizer exists. The risk-feasible subset is convex and the finite-scenario problem is linear. These assumptions retain the original multi-crop architecture while making its domain explicit.

Operational feasibility can force universal reversal without a risk mechanism. If $s_i > s_j$ and

$$\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j, \quad (5)$$

then every feasible allocation reverses the pair; $u_i + \tau_x < \ell_j$ is a transparent sufficient certificate. Conversely, if an unconstrained expected-profit optimum is risk-feasible, the risk-constrained optimal value is unchanged and the constrained optimum is the intersection of the unconstrained optimal set with the risk-feasible set. One observed slack solution does not prove equality of solution sets when optima are multiple.

The complete first-order characterization blocks a second shortcut. At an optimum x^* , non-negative multipliers and a subgradient $d \in \partial r_\alpha(x^*)$ satisfy

$$0 = -\mu + \lambda d + \gamma \mathbf{1} + \beta c + G^\top \eta - a + b, \quad (6)$$

together with primal feasibility, dual feasibility and complementary slackness. For crops i and j ,

$$\mu_i - \mu_j = \lambda(d_i - d_j) + \beta(c_i - c_j) + (G^\top \eta)_i - (G^\top \eta)_j - (a_i - a_j) + (b_i - b_j). \quad (7)$$

The common land multiplier cancels, whereas budget, shared-constraint and bound terms remain. Equation 7 describes local marginal balance; it does not rank acreage levels.

Dependence comparative statics require a distributional premise. If fixed marginals within a named family give $L_{\theta_1}(x) \preceq_{\text{cx}} L_{\theta_2}(x)$ for every allocation in a declared domain, loss-CVaR is weakly ordered there. When the domain is all of X , risk-feasible sets are nested and optimal expected profit is weakly non-increasing. No crop-specific acreage ordering follows without further uniqueness and selection assumptions. Similarly, possible and universal reversal regions defined by the maximum and minimum optimal-face acreage gaps may be empty, disconnected, multiple or interval-valued. A unique dependence threshold requires continuity, strict monotonicity and an endpoint crossing; it is not generic.

The nested M0–M4 audit makes these distinctions computationally inspectable (Fig. 3). M0 begins with a closed benchmark; successive stages add margins, operational restrictions, downside risk and dependence. Across the closed-domain seeds, the operations block shifts corn share downward and soybean share upward, whereas dependence and risk contributions have intervals that cross zero for crop shares. The all-subset Shapley decomposition closes to a maximum efficiency residual of 1.42×10^{-14} . These are exact accounting contributions over registered block orders, not causal effects.

The local pressure audit then evaluates Eq. 7 in E2. Margin pressure is balanced by budget, shared-constraint or boundary terms according to the intervention class, while the tail-risk term is zero in those cells. The maximum registered stationarity residual is 1.82×10^{-11} . This connects the theorem to a positive operational mechanism without attributing acreage shares to an omitted private farm process. Full proofs, assumptions, optimal-face constructions and order-sensitivity ranges are in the Supplementary Information.

5 Confirmatory mechanism experiments

Six experiments were frozen before execution, each with named estimands, family-wise intervals, sequential checks and a maximum replication ceiling. The resulting evidence graph contains 2320

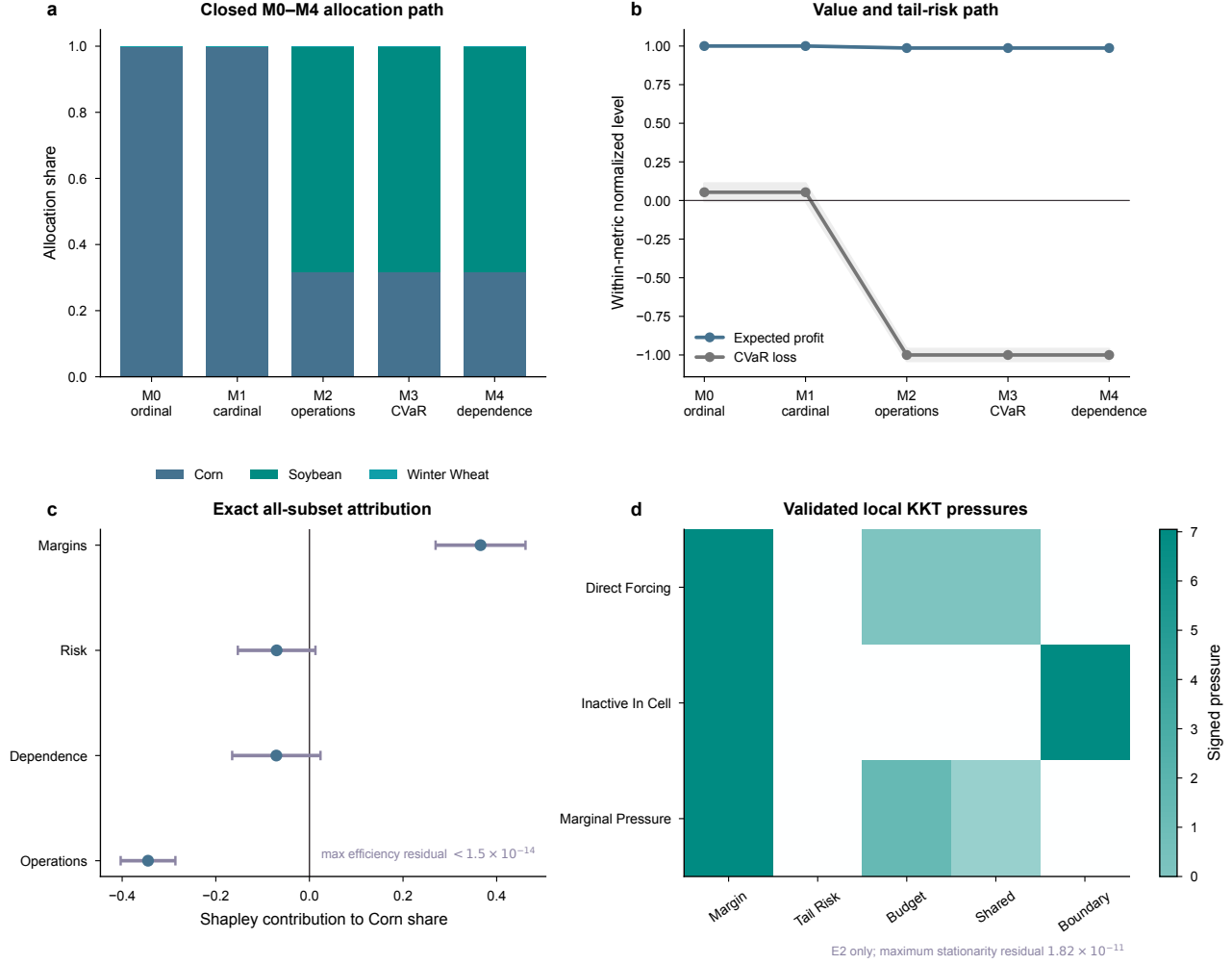


Figure 3: Nested model path and closed mechanism attribution. Panels a–c summarize 16 closed-domain seeds with descriptive 95% t intervals where shown. All-subset attribution closes with maximum efficiency residual 1.42×10^{-14} . Panel d reports validated E2 local KKT pressure terms (maximum stationarity residual 1.82×10^{-11}), not causal acreage shares.

raw rows, 5360 contrasts and 1632 scenario-registry rows. Promotion is experiment-level: every registered primary interval must meet its precision target. Under this rule 2 experiments pass and 4 fail. Feasible-only filtering is prohibited; all 206 registered infeasible rows remain in the adverse ledger.

5.1 Operational interventions identify reversal

E2 isolates budget, rotation, contract and dominant-crop-bound interventions while holding the ranking and scenario law fixed (Fig. 4). The unconstrained base assigns share 1.000 to corn. Every treated cell instead assigns a soybean majority: contract or rotation interventions produce corn/soybean shares 0.450/0.550, while the tighter budget produces 0.318/0.682. Possible, selected and universal reversal equal one in every treated cell, including the bound anchor.

All 24 of 24 registered family-wise intervals pass at 16 replications. Allocation L_1 changes are 1.10 or 1.365 relative to the base, and the selected-reversal change is one in each contrast. Because the intervention is assigned by design and the risk term is inactive in the audited cells, the experiment

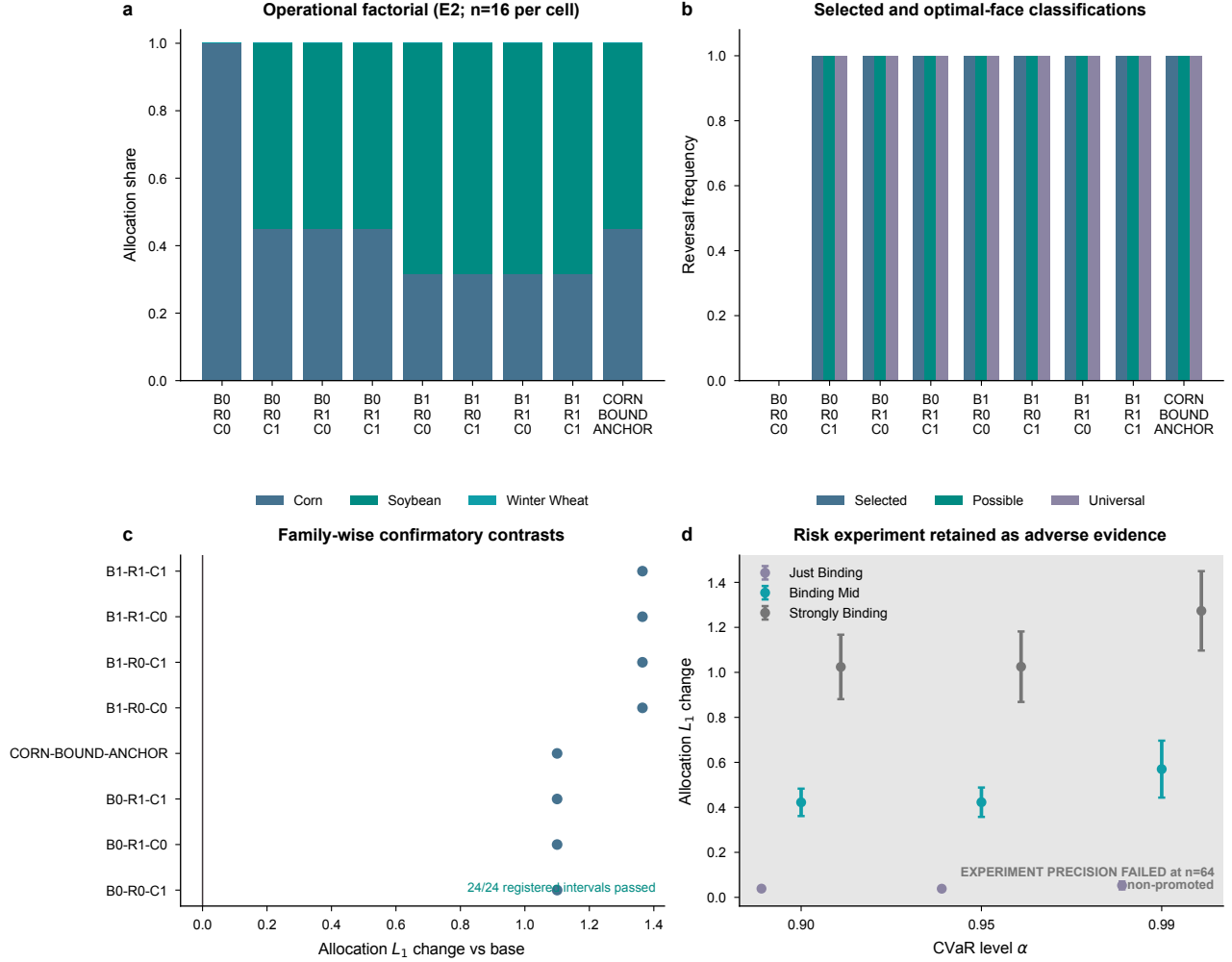


Figure 4: Operational interventions identify reversal in E2. Each E2 cell has $n = 16$; 24/24 registered family-wise intervals pass. E3 risk-frontier results are displayed as adverse evidence only because its experiment-level precision gate fails at the $n = 64$ ceiling.

identifies an operational displacement mechanism within its declared simulation domain. It does not estimate the prevalence of that mechanism in observed agriculture.

E3 asks a different question: whether moving along a CVaR frontier changes allocation and reversal. Some individual contrasts appear numerically sharp, but the experiment-level gate fails at its 64-replication ceiling. The full E3 inventory is therefore shown as adverse evidence in Fig. 4 and Supplementary Fig. S2; it is not used to claim an empirical risk effect or a general monotone frontier.

5.2 Information value depends on what flexibility enables

E6 tests a quadrant interaction between signal information and feasible-action flexibility. Define the interaction as the objective-value difference-in-differences between informative versus uninformative signals under flexible versus restricted action sets. Ignoring a signal remains feasible, so information value is non-negative in each action set; exact ignore-signal and garbling checks verify that implementation invariant.

The interaction has all three theoretically admissible signs (Fig. 5). When flexibility unlocks state-contingent specialization, the estimate is 3.258 with 95% family-wise interval [2.788, 3.728]. When

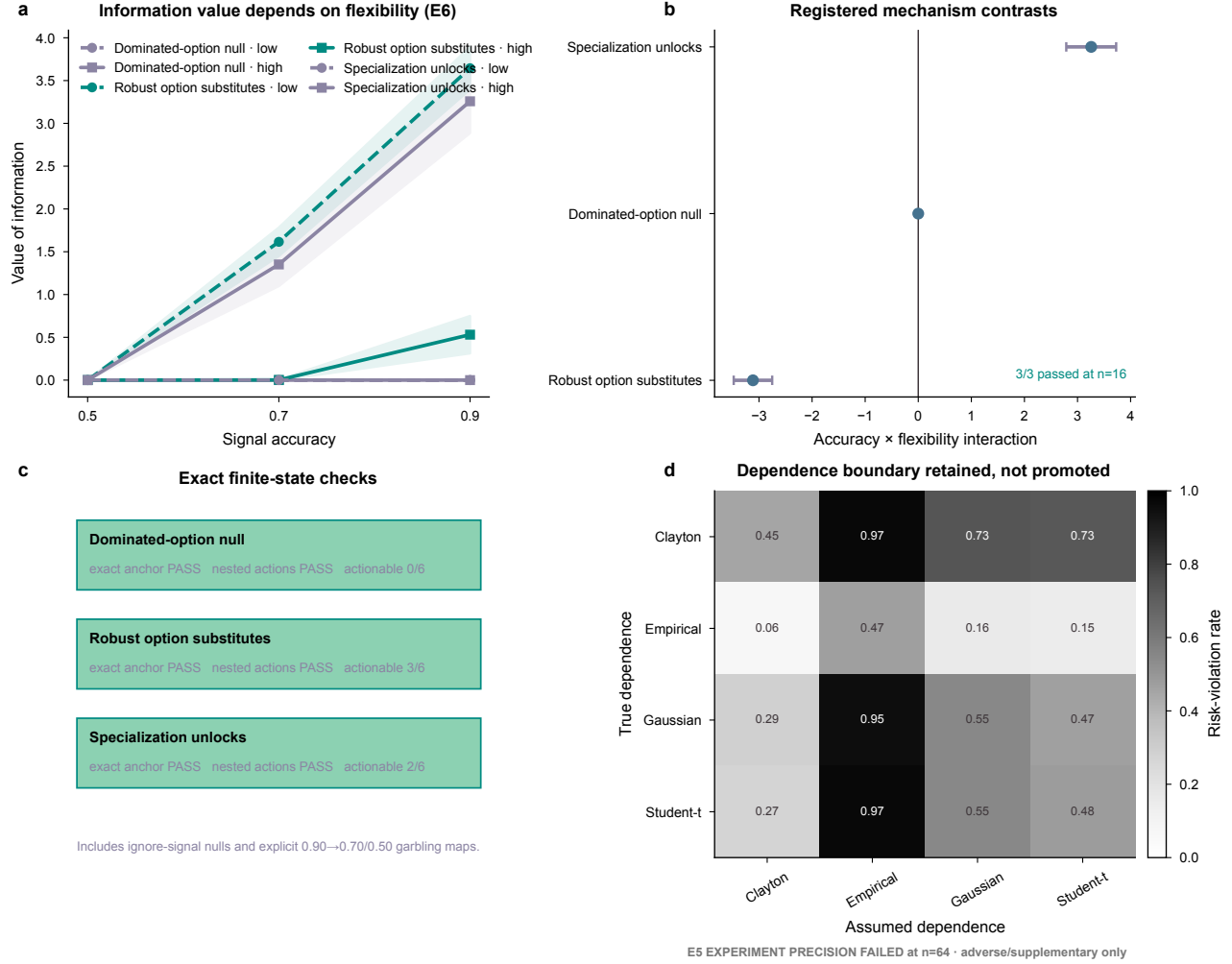


Figure 5: Information–flexibility interaction is positive, null or substitutive. E6 reports three registered interaction estimands ($n = 16$ each) with exact ignore-signal and garbling checks. E5 cross-law diagnostics remain non-promoted because the experiment-level precision gate fails at $n = 64$.

flexibility adds a robust fallback option, the estimate is $-3.114 [-3.476, -2.751]$, because flexibility is already valuable without the signal. When the added option is dominated, the interaction is exactly 0.000 . All three intervals pass at 16 replications.

This sign pattern is stronger than a single positive example: it shows why information and flexibility cannot be assumed to be universally complementary. Information is actionable when it changes the state-contingent decision, but the incremental value of the signal depends on the alternatives that flexibility adds. E5 dependence-misspecification contrasts remain adjacent as adverse evidence. Their experiment-level precision gate fails at 64 replications, so they cannot support a promoted claim about copula misspecification, regret or downside risk.

5.3 Adverse evidence and numerical integrity

E1, E3, E4 and E5 reach the registered ceiling without satisfying every primary precision interval. Their estimates, intervals, stopping reasons and infeasible cells remain visible in Supplementary Figs. S1 and S2. Failed experiments are not pooled with E2 or E6, and passing sub-contrasts do not rescue a failed experiment. This rule prevents selective promotion after looking at the results.

Numerical diagnostics support implementation integrity across both promoted and adverse experiments. Reverse-order replay passes 12/12 checks, solver sensitivity passes 9/9, the maximum KKT residual is 1.82×10^{-11} , and the maximum Shapley efficiency residual is 1.42×10^{-14} (Supplementary Fig. S3). These checks show that the recorded results are reproducible and internally consistent; they do not turn imprecise estimands into evidence.

6 Data and empirical design

The empirical analysis uses the admitted official crop-production summary for state planted acreage and yield in 2022–2024, together with admitted national price, cost and CPI series. We retain state-years with complete acreage and yield for corn, soybeans and winter wheat, without imputing published missing codes. The final panel contains 231 crop rows, 77 state-years and 26 states. It is a complete-case aggregate panel, not a nationally representative sample of farms.

Four score definitions are frozen before comparison with acreage. Relative yield divides state yield by the same-year national yield. Standardized revenue multiplies state yield by the national same-year price. Operating margin subtracts national operating cost per planted acre, and total-cost margin subtracts national total cost. Monetary variables are expressed in real 2024 US dollars using CPI-U. These definitions are transparent accounting constructions: they are not local realized margins, farmer objectives or welfare measures.

For each state-year and definition, crops receive descending score ranks and planted-acreage shares. Concurrent pairwise inversion intensity is the fraction of informative crop pairs for which the score and acreage orders disagree; ties are retained explicitly. Top-rank reversal records whether a unique score leader is not the acreage leader. We use 5000 deterministic state-cluster bootstrap draws for 95% intervals, preserving within-state dependence across the three-year panel.

The temporal design creates two rolling-origin transitions per eligible state and score definition, giving 51 transitions per definition, 204 rank-transition events and 612 crop-transition rows. Scores at $t - 1$ are compared with acreage-share changes from $t - 1$ to t . The top-minus-other contrast asks whether the previously top-scoring crop changes share differently from the other two crops. Prior acreage enters only as an inertia proxy. It is not labelled a rotation, capacity or adjustment constraint.

Aggregation is predeclared as an identification boundary. State-level results are compared with national crop-year ranks under the same definitions. Relative-yield national scores are tied by construction, so those comparisons are non-informative rather than treated as agreement. Two isolated raw-to-output executions reproduce all 19 empirical artifacts byte for byte. Source identifiers, timestamps, units, raw checksums, processing contracts and parameter scope are registered.

The admitted sources do not reveal private budgets, rotations, contracts, risk limits, beliefs or objectives. Observed acreage therefore cannot be certified as an optimum of the stochastic programme. The empirical estimands describe score–acreage order relations and transitions; they do not identify CVaR, copula causality, welfare or a private constraint mechanism.

7 Empirical patterns and aggregation boundaries

Concurrent score and acreage orders differ materially in the admitted state sample (Fig. 6). For operating margin, inversion intensity is 0.597 with a state-cluster 95% interval of 0.494–0.697; the score leader differs from the acreage leader in 81.8% of state-years. The corresponding inversion intensities are 0.706 for relative yield, 0.351 for standardized revenue and 0.680 for total-cost margin. Thus rank disagreement is visible but inherits the definition of the ordinal object.

State and year panels show that the aggregate is not spatially or temporally uniform. Leave-one-state-out operating-margin inversion intensity ranges from 0.586 to 0.617, while yearly estimates range

from 0.547 to 0.692. These summaries are descriptive: the panel is short, the state sample is selected by three-crop completeness, and the cluster intervals quantify finite-sample stability under the frozen resampling design rather than causal uncertainty.

The lagged transition test does not provide positive evidence that a previously higher score predicts a larger subsequent acreage-share change. For operating margin, the top-minus-other contrast is 0.0089 with interval $[-0.0018, 0.0213]$. The analogous intervals for relative yield, standardized revenue and total-cost margin also include zero. Across definitions, the score leader can change substantially more often than the acreage leader; the latter changes in only 5 of 51 state transitions. This is consistent with acreage inertia, but it does not identify the source of inertia.

National aggregation changes the pattern. Operating-margin and standardized-revenue inversion intensities are both 0.000 nationally, whereas total-cost-margin inversion intensity is 0.667. Relative yield produces 9 pairwise national ties and therefore no informative national comparison. The national results neither refute state heterogeneity nor validate a particular score. They show that the spatial support of a ranking-reversal statement must be declared.

The combined empirical result is deliberately asymmetric. Concurrent inversions are frequent in this panel, but a temporally ordered response of acreage shares is unresolved and national patterns depend on the accounting definition. Public aggregate data therefore document a phenomenon compatible with the model's ordinal-cardinal distinction without selecting among margin, operational, risk, dependence or multiplicity mechanisms.

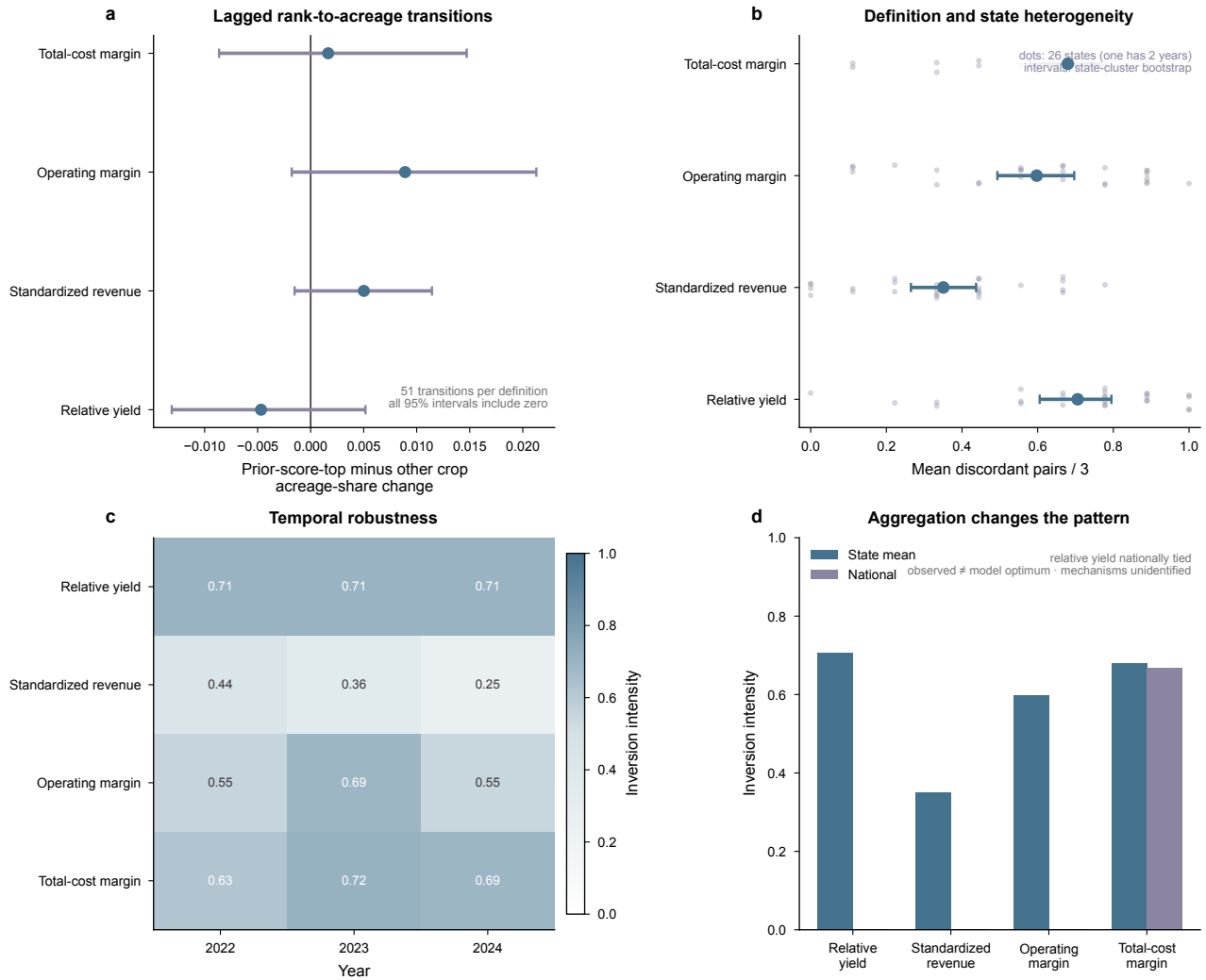


Figure 6: Admitted-data heterogeneity and aggregation boundary. Evidence covers 77 state-years in 26 states (2022–2024). Across 51 transitions per definition, every 95% state-cluster interval for the prior-score-top versus other-crop mean share change includes zero. Concurrent inversion intensity varies by definition, state, year and national aggregation. Relative yield is nationally tied. No private-objective, constraint, CVaR, copula, causal, welfare or optimal-acreage inference is made.

8 Robustness and extensions

Three robustness layers are kept distinct. Leave-one-state-out and year-specific summaries test whether the concurrent inversion intensity is mechanically driven by a single cluster or crop year. Rolling-origin transitions impose temporal ordering but remain descriptive because past public scores are not farmer beliefs. National aggregation changes the unit of analysis and, for relative yield, creates ties by construction. None of these checks supplies the missing private feasible set.

The simulation has an equally explicit robustness hierarchy. Numerical replay, solver-method agreement, KKT closure and Shapley efficiency are implementation checks. Registered intervals determine inferential precision. The E2 result is additionally robust to separate and combined budget, rotation, contract and corn-bound interventions within the frozen cell family. E6 includes constructive positive, exact-null and substitution cases, so its conclusion does not rest on assuming that every added action is useful in every information state.

The framework can accommodate stochastic costs, fixed charges, endogenous prices, recourse after information, spatial coupling and multi-period rotations. Each extension changes the decision problem and requires new assumptions or data. In particular, a dynamic acreage model could separate physical rotation from adjustment inertia, while farm-level observations on expectations and contracts could connect controlled operational mechanisms to field decisions. Named dependence laws could be compared under verified portfolio-loss orders, but the present failed E4/E5 gates do not authorize a general dependence claim.

9 Discussion

The central result is that ranking reversal is not one mechanism. It is a relation between an ordinal statement and a cardinal allocation drawn from a constrained optimal set. Exact theory identifies when ordinal information is insufficient and when feasibility alone forces reversal. Precision-qualified experiments then demonstrate two positive mechanism classes: operational interventions can overturn a fixed ranking, and the value interaction between information and flexibility changes sign with the actions that flexibility makes available.

This mechanism view improves on both extremes of interpretation. Treating every score-acreage mismatch as an error ignores objectives, constraints and uncertainty. Treating every mismatch as uninformative misses controlled cases in which the intervention and optimal face are known. E2 identifies operational displacement because the constraint intervention is assigned inside a closed simulation. E6 identifies actionability because signal and action-set contrasts are factorial and invariant checks pass. Neither result makes those mechanisms empirically prevalent.

The failed experiments are scientifically informative. E1, E3, E4 and E5 show that deterministic reproduction, solver agreement and visually suggestive contrasts do not guarantee registered precision. They remain nonheadline and cannot support a unique dependence threshold, a general CVaR effect or a copula-misspecification penalty. Retaining all failures and infeasible rows makes the evidence rule inspectable rather than retrospective.

The empirical evidence establishes a different layer. Concurrent rank-acreage inversion is common in the complete-case state panel, yet it is definition-sensitive; lagged share-change contrasts include zero; and the national comparison ranges from null to two-thirds depending on definition, with relative-yield ties. The optimizing mechanism is not identified. In particular, the data provide no private constraints or objectives, and they do not identify CVaR preferences, copula causality, welfare, optimal acreage or why an individual state-year differs.

Several limitations follow. The empirical panel covers three crops and three years and uses national price and cost inputs to standardize accounting. Observed acreage embeds unobserved expectations,

policies, contracts, rotations and adjustment costs. The baseline model is linear and single-period, omitting fixed costs, endogenous prices, spatial interaction and dynamic rotations. The promoted simulations demonstrate existence and sign heterogeneity within declared domains, not population frequencies. Their parameters are controlled design values rather than farm estimates.

The practical implication is procedural. A ranking system intended to guide acreage should report the cardinal mapping from scores to payoffs, the provenance of operational restrictions, the joint uncertainty model, the risk convention, the optimizer-selection rule and sensitivity to plausible alternatives. A signal is valuable only if it can alter a feasible action, and added flexibility can complement, substitute for, or be irrelevant to that signal. When decision inputs are unavailable, rank-acreage comparisons should be labelled descriptive rather than converted into efficiency or compliance judgments.

10 Conclusion

Crop rankings and land allocations are different mathematical objects. A set-valued stochastic programme shows why: cardinal margins, joint uncertainty, operational feasibility and optimizer selection all intervene between an order and an acreage vector. Operational constraints can force universal reversal, while dependence and information claims require stronger conditions than a scalar index or a general appeal to flexibility.

The Stage II evidence supplies positive results without erasing boundaries. E2 identifies operationally induced reversal in all treated cells, and E6 shows positive, null and substitutive information–flexibility interactions. Four other experiments remain adverse because their precision gates fail. Admitted state data reveal concurrent ranking-acreage inversions but no robust lagged share response and no aggregation-invariant rate. Ranking reversal is therefore demonstrable under explicit mechanisms and empirically visible as a descriptive pattern, but its field mechanism must be measured rather than inferred from disagreement alone.

11 Methods

11.1 Theory and optimization

Every preserved theorem was classified before implementation. The canonical model uses portfolio loss rather than the favourable profit tail, permits general positive finite-scenario weights, leaves the VaR auxiliary variable free and retains land, budget, shared-constraint and bound multipliers. Possible and universal reversal are computed by minimizing and maximizing pairwise acreage gaps over the objective-equivalent optimal face. Equality, feasibility and complementarity checks use frozen tolerances.

11.2 Confirmatory design and computation

The six-experiment design was frozen before execution. Each experiment specifies factors, estimands, multiplicity, sequential replication checks, precision targets and a ceiling. E2 crosses operational interventions against a fixed ranking and law. E6 crosses signal information and feasible-action flexibility in three constructive environments. E1, E3, E4 and E5 test margin, risk-frontier and dependence questions. Scenario seeds, laws and hashes are registered. Each solve records status, objective, allocation, loss-CVaR, named slacks, binding flags, duals, tail weights, KKT residuals and optimal-face gaps.

Family-wise intervals use the coverage and multiplicity frozen for each experiment. Promotion requires every registered primary interval to pass at the experiment level; isolated passing contrasts

inside a failed experiment remain adverse. Registered infeasible rows are preserved and classified, never discarded to form a positive subset. Reverse-order replay reconstructs selected tasks, and solver sensitivity compares HiGHS methods at registered anchors.

11.3 Mechanism attribution

The M0–M4 path adds registered blocks to a closed benchmark. Because sequential contributions depend on order, all subset coalitions are solved and Shapley contributions are computed for allocation, expected profit, loss-CVaR and concentration. The efficiency residual compares the sum of contributions with the full-minus-base difference. KKT pressure terms are evaluated locally from Eq. 7; they are labelled accounting terms, not causal acreage effects.

11.4 Official data and empirical estimands

Raw sources enter only through a registry containing authority, stable identifier, retrieval date, geography, time support, units and checksum. The parser applies a complete three-crop rule. National prices and costs are joined after crop-year reconciliation and converted to real 2024 dollars. Four ranking definitions are computed without post-result selection.

Concurrent inversion intensity uses informative pairwise comparisons within state-year. Cluster intervals resample states with replacement for 5000 deterministic draws. Rolling-origin contrasts use the previous year’s score status and the subsequent change in planted-acreage share; state clustering preserves serial and cross-crop dependence. National aggregation repeats the ranking construction and marks tied definitions non-informative. All empirical claims are descriptive or accounting-identified.

11.5 Visualization, reproducibility and claim governance

Figure scripts read only verified theory, simulation and empirical outputs. The frozen colour card is #3D3539, #0F9EA8, #008B82, #45728F, #8CD1B2 and #8B84A3; colour is redundant with text, position, marker or line style. Every figure has SVG, PDF, 300-dpi PNG and 600-dpi TIFF exports, tidy source data, checksums, grayscale proofs and deuteranopia simulations.

Manuscript numbers are generated from verified outputs and registered at field level. Claim–citation and figure-use registries link prose to theory, simulation, empirical evidence and source data. The paper and Supplementary Information are compiled twice in isolated directories with a fixed source date; byte identity, embedded fonts, page rendering, references and checksums are validated before packaging.

Declarations

Acknowledgements. To be completed by the authors before submission.

Author contributions. To be completed using the journal’s CRediT format.

Competing interests. The authors must confirm the final statement before submission.

Ethics. The study uses public aggregate data and no human-participant or animal data.

Data availability. All admitted source identifiers, immutable raw-file checksums, processing contracts and derived analysis files are recorded in the repository’s data and evidence registries. Public-source redistribution remains subject to the originating agencies’ terms.

Code availability. The complete theory, optimization, simulation, empirical, visualization and manuscript pipelines are versioned in the companion repository. Reproduction commands and checksums are provided with each artifact class.

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